

Grey level transformation :-

→ Grey level transformation is used to improve the appearance of an image by changing the brightness of its pixels. A positive transformation highlights light areas as dark and dark areas as light.

formula $s = 255 - r$

original pixel

50

100

150

200

Transformed pixel(s)

205

155

105

55

* Therefore, the image becomes a positive image. Bright regions become dark, and dark regions become bright.

2. Histogram equalization :-

→ Histogram equalization improves image contrast by spreading pixel intensity values over a wider range.

Cummulative distribution function (CDF) :-

Intensity

0

1

frequency

4

6

CDF

4

10

<u>Intensity</u>	<u>frequency</u>	<u>CDF</u>
2	10	20
3	6	26

∴ Total pixels = 26

Equalized Intensity value :-

$$\text{New Intensity} = (\text{CDF}/26) \times 3$$

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<u>Intensity</u>	<u>CDF</u>	<u>New intensity</u>
0	4	0
1	10	1
2	20	2
3	26	3

* Therefore, the pixel values are more evenly, making the image cleaner with better contrast.

3. Gaussian Smoothing :-

→ Gaussian Smoothing reduces image noise while keeping edges smoother than an average filter.

Given,

gaussian kernel $\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$

image region $\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$

$$= (10 + 20 + 10 + 40 + 160 + 40 + 10 + 40 + 10) / 16$$

$$= \frac{360}{16} \approx 22.5 \approx 23$$

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<u>original value (x)</u>	<u>calculation</u>	<u>New values</u>
50	$(50 - 50) \times 255 / 100$	0
100	$(100 - 50) \times 255 / 100$	128
150	$(150 - 50) \times 255 / 100$	255

* Therefore, the intensity range changes from (50-150) to 0-255 making the image brighter and improving its overall contrast.